

Agricultural Supply Chain Management Using Blockchain Technology

Bhagya Hegde
IEM department
BMS College of Engineering
Bengaluru, India
bhagyarh813@gmail.com

Dr. B Ravishankar
IEM Department
BMS College of Engineering
Bengaluru, India
ravi36@gmail.com

Mayur Appaiah
IEM department
BMS College of Engineering
Bengaluru, India
mayurappaiah.iem@bmsce.ac.in

Abstract—In this day and age, agricultural producers are faced with multiple obstacles, from seasonal changes to the broken supply chain; their occupation is very laborious and demanding. In such a situation, a distinct information database consisting of credible information would be very helpful. Transfer of knowledge is important in all aspects of this occupation, may it be about market trends or about profitable practices. Third-party interference in this aspect may cause the spread of misinformation. This can be curbed by using blockchain, a data ledger which is reliable and incorruptible. Here an attempt to analyse the different ways in which blockchain technology can be incorporated in the agricultural supply chain, as a transparent and dependable transaction mechanism is explored.

Keywords—Agricultural supply chain, Blockchain, Information database, Resource wastages.

I. INTRODUCTION

Blockchain is a computing technology which facilitates the transfer of data or useful information in a decentralized and transparent manner. It was invented by Satoshi Nakamoto in 2008, and it serves as a distributed ledger that verifies, records and timestamps transactions. Don and Alex Tapscott, authors of “*Blockchain Revolution*”, describe it as “an incorruptible digital ledger of economic transactions” (Don & Alex Tapscott, 2016)[1]. It is an ever-growing field of technology, with applications in industry, manufacturing, logistics, agriculture and also the government sector (Yli-Huumo, Jesse & Ko, Deokyoony & Choi, Sujin & Park, Sooyong & Smolander, Kari, 2016)[2].

Blockchain also includes smart contracts. They are computable contracts or protocols which are previously agreed upon by the different parties and are enforceable automatically without any human intervention. Different parties who might not trust each other will be able to undertake business with each other using smart contracts. Thus it reduces moral hazards in the industry. The working diagram is shown in Fig. 1.

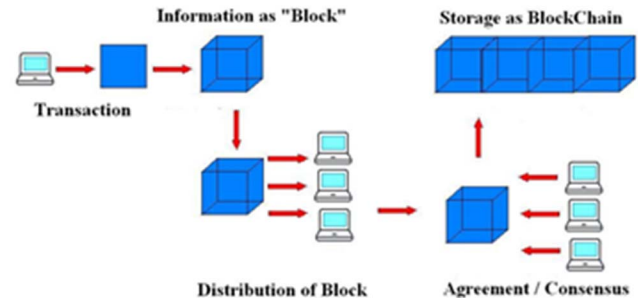


Fig. 1. Working of Blockchain.

There are multiple technical challenges this technology is facing currently regarding its security, usability, wasted resources and other aspects (Al-Jaroodi, Jameela & Mohamed, Nader, 2019)[3]. However, further research is required before one can find solutions and improve upon it. Many regulations also have to be created.

II. EXISTING AGRICULTURAL SUPPLY CHAIN

The existing agricultural supply chain in India is highly fragmented. It consists of multiple agents or intermediaries tapping into the marketing channel to realize a profit and successfully pass on the losses to the producer. It is an age old system which has gone through years and years of customisation. Owing to all the market research done over the years, the system is highly efficient. It works because everybody knows where they stand.

As shown in the figure.2, the farmers may sell to traders, both retailers and wholesalers. This depends on the type of produce grown. Example: commercial crops such as coconut and areca nut are sold to wholesalers or institutions like APMCs. Horticultural crops such as pineapples and other fruits are sold to retailers. Also, there exist secondary processors who will utilize the produce to create a product of their own. Example: vanilla, cocoa. On failure of sale through other mediums, producers may sell the produce directly to the customers.

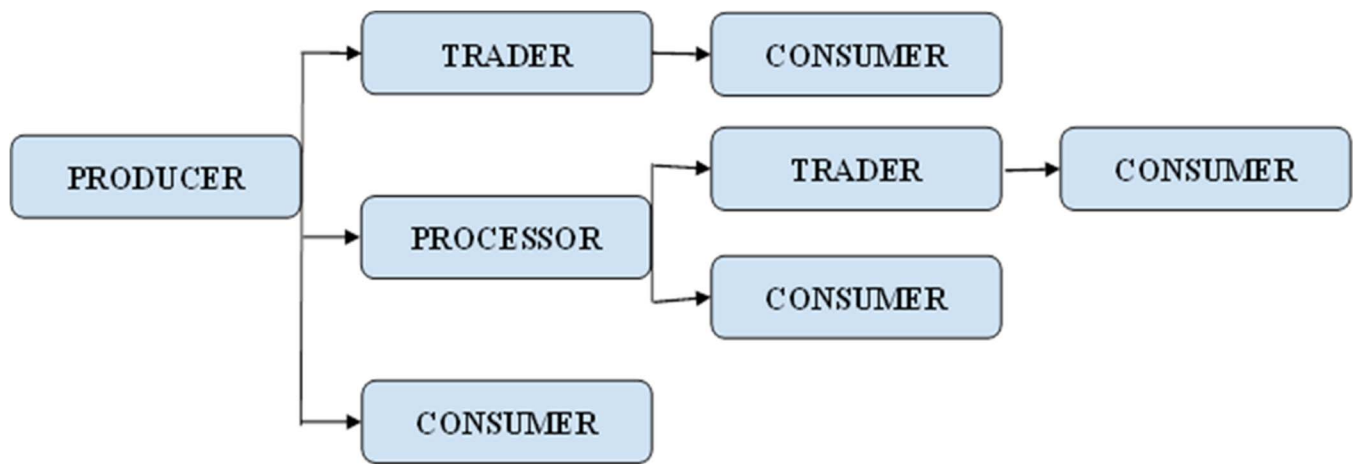


Fig. 2. Linear Agricultural Supply Chain.

As mentioned before, the existing system is efficient (i.e., the traders make profits and the customers get goods). Even then around 16% of fruits and vegetables, and around 10% of oilseeds, pulses and cereals are wasted every year in India (“India wastes 16% of produce”, 2019)[4]. It is mainly because of the lack of communication between the different tiers of the supply chain. Most producers, smallholders in particular, have limited knowledge about markets, pricing, and quality control and in general are uninformed. The market research done by the higher tier players of the supply chain who sell directly to the consumers is not shared with the lower tiers. Thus the production and transfer of inventory is not efficient and leads to wastages.

For example, let's consider the perishable goods section. There are a lot of price and demand fluctuations during the sale of these goods. The prices see a dip during the harvest season since there is an abundance of produce. Though there is a high demand, wastage can be observed. But the price soars during the off season. Thus goods stored in cold chains are sold with high profits. But the demands may not be met since there is not enough inventory to be sold. This is due to the uninformed decisions made by the lower tier players/farmers who did not invest in or sell to cold chain logistics. Thus, the lack of transfer of information has been the main cause of this shortcoming.

III. BLOCKCHAIN APPLIED TO AGRICULTURAL SUPPLY CHAIN

Since the supply chain is so fragmented, the transfer of credible information is slow and many times not to be found. Data about market trends, seasonal changes, pricing, quality and quantity requirements does exist, but is not readily available to the producer in a customized

and easily comprehensible format. APMCs upload the detailed account of sale of different crops in different places on a day to day basis to their website (Karnataka State Agricultural Marketing Board [KSAMB], 2019)[5].

The India Meteorological Department, updates weather forecasts and warnings. Different processors and traders undertake market research. There are numerous data sources which have information about different concerns regarding this logistics network. Blockchain can be applied in this facet. The data available to any player in the network must be highly reliable. Thus using blockchain will make sure the data is not manipulated at any step and is time stamped. Miners and forgers via different nodes will be able to process data from every tier of the network and provide customized data sets. This will ensure transparency and help every individual make informed decisions in favor of both personal interest and the bigger picture. For this to work, all parties must agree and deliver their part. Since the network is very fragmented and there are multiple individuals/companies working towards a personal goal, smart contracts can be utilized. Usage of smart contracts will ensure that nobody undergoes abuse and everybody can work towards achieving their personal goals without marginalizing any one tier.

Consider the farmer's protest against the sugar factories in Karnataka. The Karnataka Sugarcane (Regulation of Purchase and Supply) Act, 2013 states that the occupier of the factory shall make the payment of additional sugarcane price within fourteen days from the date of declaration[6], yet the factories owed the farmers around 3990 Crore Rupees as of 2018-19 (“In Karnataka's Belgaum, sugarcane farmers are being pushed to the edge”, 2019)[7]. A smart contract signed between the producers and the factory units would eliminate this problem.

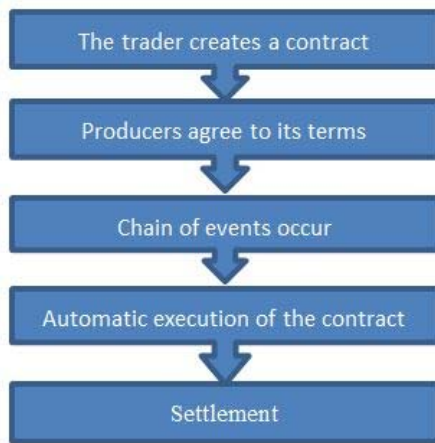


Fig. 3. Basic Working of Smart Contract.

As shown in fig. 3., a contract is developed. All the terms and conditions of the contract must be agreed upon by both the parties involved. Once the aforementioned events occur the contract executes automatically and the business is concluded.

But the occurrence of this situation is due to multiple factors. There exists a huge supply and demand gap in the Indian sugar industry. India produced 33.16 million tonnes of sugar in 2018-19 but consumed close to only 24.6 million tonnes (“Sugar production set to fall 21.6%”, 2019)[8]. This situation in the domestic market causes decrease in sugar prices affecting profitability. Thus this oversupply of sugar has caused a downward trend in sugar production which is set to continue in the future. Furthermore, poor water management and over exploitation of soil among other reasons are causing decrease in yield of the crop per acre (“India’s sugar output could drop 15% as drought hits”, 2019)[9]. Most of the sugarcane farmers follow monoculture, thus they depend only on this specific crop for their livelihood. They are slow in adapting to new farming techniques and are also unaware of factors that affect both the production and demand of their yield. Thus they incur net losses.

Most of the losses produced here are due to misinformation, miscommunication and lack of trust between the different tiers. Introduction of blockchain can cause the fragmented nature of this supply chain to be simplified. Consider a database consisting of credible information regarding the various aspects of this supply chain, as shown in fig. 4. This database is integrated with third party systems such that reliable information is constantly being included. The use of blockchain implies that all the data is time stamped and peer-to-peer reviewed. Utilizing this, all the participants can make well informed decisions.

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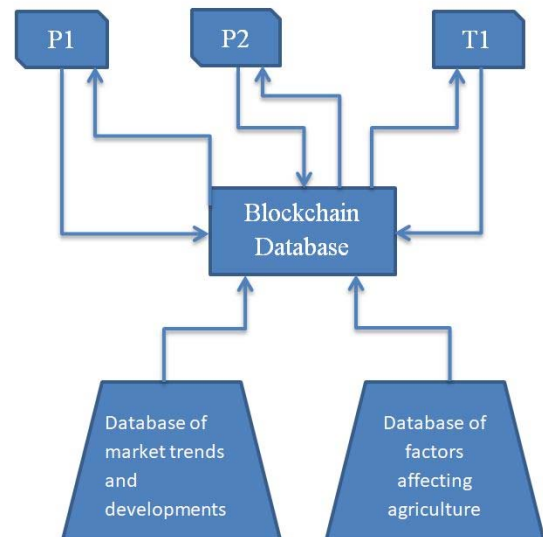


Fig. 4. Blockchain Database.

IV. ADVANTAGES

- Producers will have security. They will be able to receive a reasonable percent of profit. They will have knowledge regarding the various factors affecting their crops (i.e., unprecedented seasonal changes, soil quality, water availability, etc.) and thus will be able to take well informed decisions.
- Traders can eliminate competition.
- Processors will be able to procure goods they specifically want for the product they are trying to produce and also control the quality from the bottom up.
- Traceability: Players in different tiers can be held responsible for any damages.
- Increase in multidirectional information transfer.
- Overall increase in efficiency: Only required amount of goods will be produced thus decreasing wastages and increasing profits.

V. CONCLUSION

By employing blockchain in different aspects of the Indian agricultural supply chain, many problems found, can be cleared up. Data collection, verification, storage and transfer will be done in a transparent and reliable manner. Usage of which will lead to increase in efficiency, decrease in wastage and overall improvement in the industry.

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